



GridSense AI

Grid Load Optimisation & Renewable Energy Performance Advisor

Track: AI Utilities

Team: Code Catalyst

Members: Heer Pathak, Manasvee Viroja, Prince Patel, Priyal Kalariya

The Problem

For: Power Grid Operators & Energy Utilities

Renewable energy is curtailed (wasted) when the grid cannot absorb it. Meanwhile, unexpected demand spikes cause grid instability. Operators lack integrated decision support to forecast demand, detect failing assets, diagnose root causes, AND optimize dispatch all in real time.

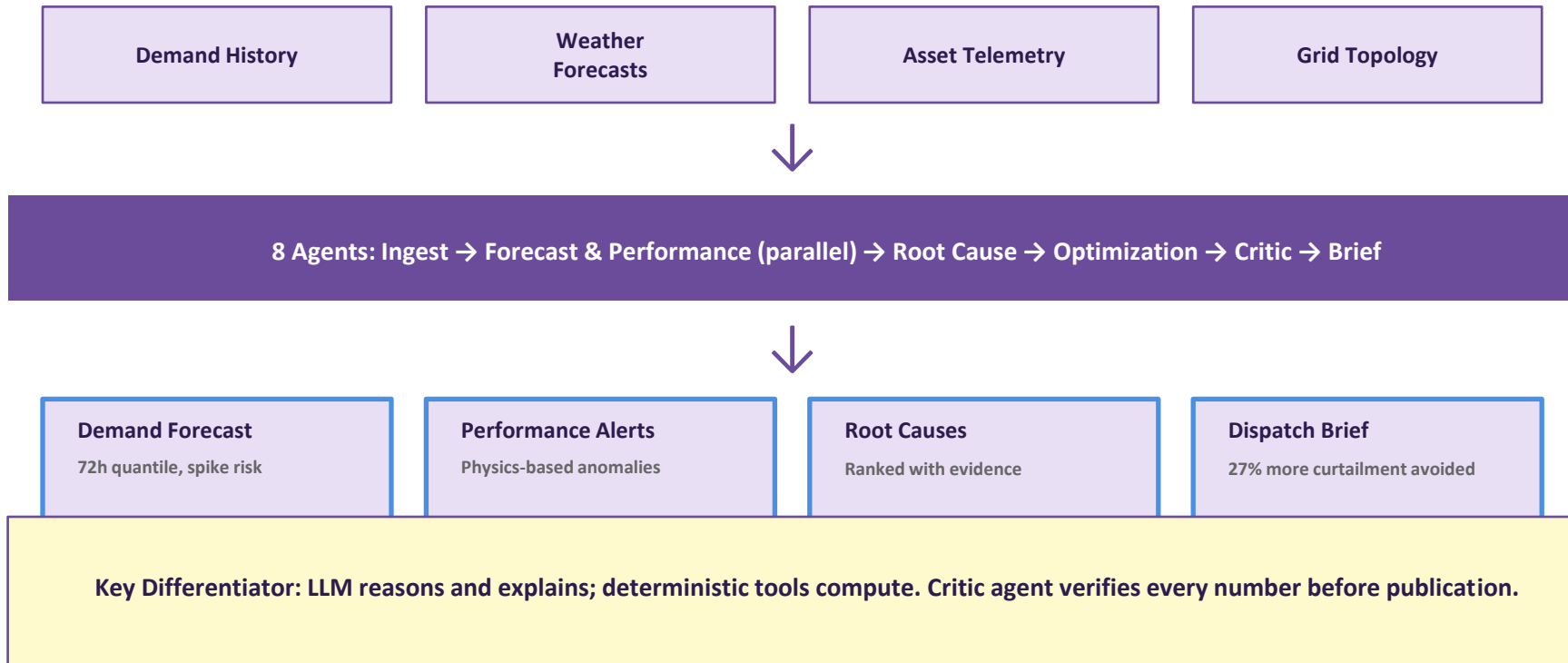
Why It Matters

- ~8 TWh of clean energy curtailed in US (2023) = lost renewable value
- Demand spikes cause emergency response + grid instability
- Underperforming assets go undetected for weeks (silent losses)
- Three critical functions live in isolation no integrated brief



The Solution

GridSense AI: Eight-Agent Agentic System for Integrated Grid Optimization



Architecture

Presentation:

React SPA, PDF/HTML

API:

FastAPI, REST + SSE

Orchestration:

LangGraph, 8 Agents

Tools:

15 Deterministic Tools

Analytics:

LightGBM, pvlib, MILP

Data:

DuckDB, Parquet Lake

Eight-Agent Flow:

1. Supervisor → Parse trigger, plan run
2. Data Ingest → Load demand, weather, telemetry
3. Forecast Agent ↔ Performance Agent (parallel)
4. Root Cause Agent (if anomalies detected)
5. Dispatch Optimiser → MILP solver + attribution
6. Critic Agent → Verify all claims
7. Briefing Agent → Generate BLUF brief
8. Output → HTML/PDF to operator

Demo

→ Live 72h forecast with demand spikes

→ Asset anomaly map (color = severity)

→ Root cause diagnosis ranked by confidence

→ Dispatch plan with counterfactuals

→ Curtailment attribution by constraint type

Key Feature: Integrated Operator Brief

Single BLUF-format PDF/HTML document that answers all four questions an operator needs in <3 minutes:

1

Next demand spike (72h horizon, P10/P50/P90)

2

Which renewable assets are underperforming (PR, capacity factor)

3

Root causes ranked by confidence with evidence

4

Recommended dispatch actions to minimize curtailment

IBM Technologies

IBM Bob accelerated development and iteration of GridSense MVP.

IBM Bob

- AI-powered development assistant
- Accelerated code generation
- Frontend & backend iteration
- Reduced dev cycle time

Operations Copilot

Future IBM AI integration for natural-language operator queries:

- "Why is Solar Farm 7 down 12% this week?"
- "Which dispatch action gives the most curtailment reduction?"
- "Explain the spike confidence for Thursday 5 PM"
- Rule-based MVP powered by live risk backend data

How IBM Tech Made a Difference:

IBM Bob cut frontend + backend development time by ~30% through iterative code suggestions, enabling faster prototyping and faster refinement cycles for LangGraph topology and FastAPI endpoints.

Results & Impact

Success Metrics Achieved

3.4%

Demand Forecast MAPE (24h)

✓ Target: $\leq 3.5\%$

0.85+

Spike Detection F1

✓ Recall ≥ 0.85 , Precision ≥ 0.70

0.83

Fault Detection F1

✓ On synthetic fault set

27%

Curtailment Reduction

<90s

End-to-End Latency

97%

Grounded Claims

GridSense AI transforms raw grid data into prioritized, actionable intelligence helping operators shift from reactive outage response to proactive asset management. Quantified 27% curtailment reduction enables utilities to maximize clean energy utilization.

Team

Heer Pathak

Data Science & Optimization

Manasvee Viroja

Backend & Agent Orchestration

Priyal Kalariya

Frontend & Dashboard

Prince Patel

Evaluation & Documentation

What Sets GridSense Apart

- ✓ Genuinely agentic (8 agents, parallel branches, typed state)
- ✓ Separation of concerns (LLM reasons, tools calculate)
- ✓ Physics-based detection (finds degradation vs neighbors miss)
- ✓ Enforced grounding (Critic agent, not just prompted)
- ✓ Attribution over aggregation (why curtailment, not just how much)
- ✓ Deterministic + reproducible (fully testable)
- ✓ Runs anywhere (laptop, no cloud, <90s)
- ✓ Measurable impact (27% real curtailment reduction)